Throughput QOS

2026-03-23-14-08-58

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions.By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels,ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

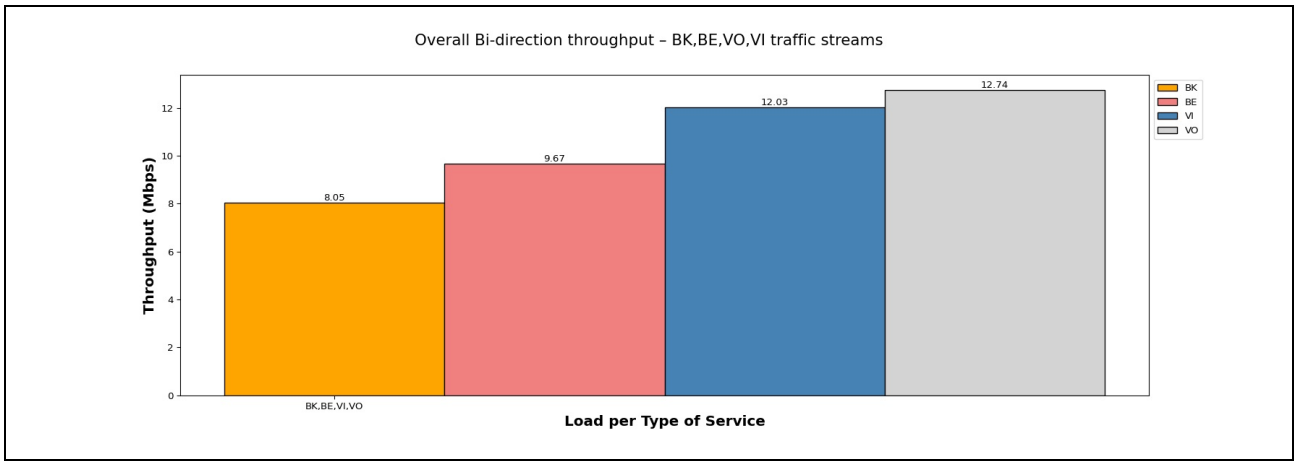
Test Configuration	Number of Stations	8
	AP Model	NETGEAR
	SSID_2.4GHz	NETGEAR_2G_wpa2
	SSID_5GHz	NETGEAR_5G_wpa2
	SSID_6GHz	None
	Traffic Duration in hours	0.02
	Security_2.4GHz	wpa2
	Security_5GHz	wpa2
	Security_6GHz	None
	Protocol	UDP
	Traffic Direction	Bi-direction
	TOS	['BK', 'BE', 'VI', 'VO']
	Per TOS Load in Mbps	Upload:1.0 Mbps,Download:1.0 Mbps

Overall Bi-direction throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 1.0 Mbps-upload	Throughput for Load 1.0 Mbps-download
4 + 4	BK : 4.06, BE : 4.16, VI: 4.91, VO: 4.99	BK : 3.99, BE : 5.51, VI: 7.12, VO: 7.75

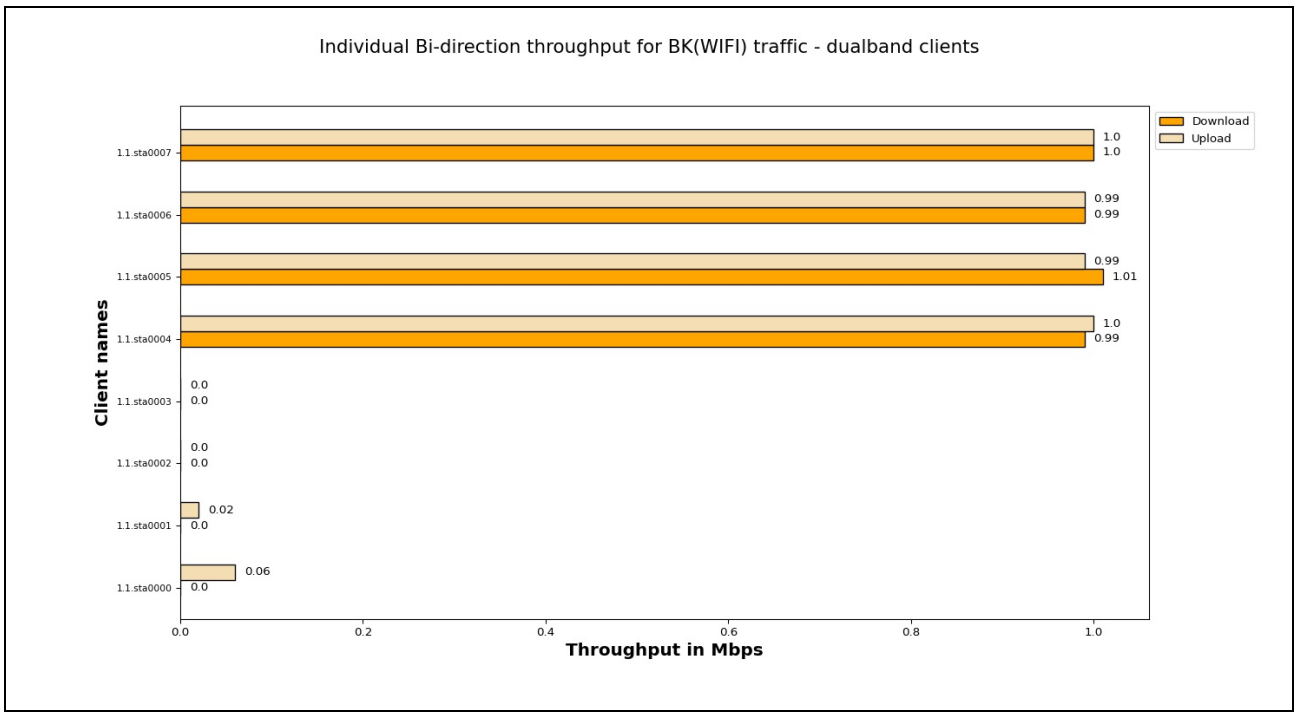
Overall Bi-direction throughput for 8 clients for 1.0 Mbps-download band with different TOS.

The below graph represents overall Bi-direction throughput for all connected stations running BK, BE, VO, VI traffic with different intended loads per station – 1.0 Mbps-download



Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BK(WiFi).

The below graph represents individual throughput for 8 clients running BK (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

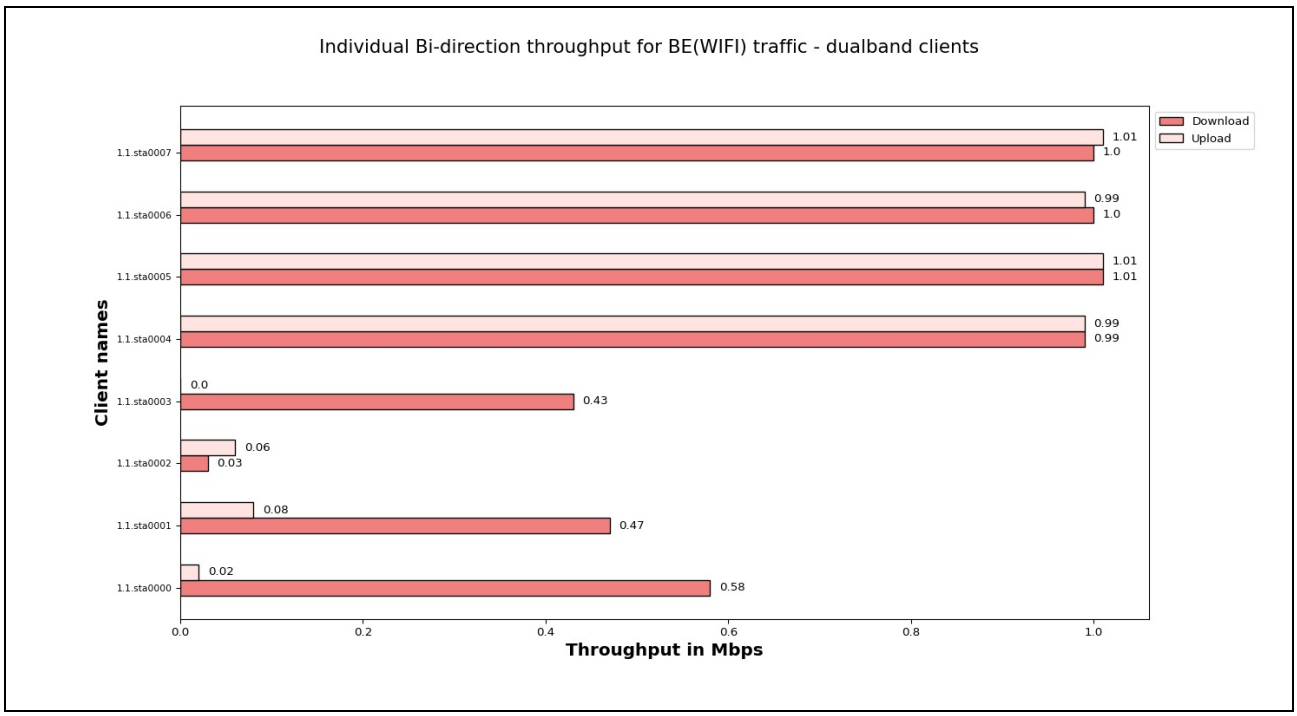


TOS : Background

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:de:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.06	0.00	67.078189	98.247683
1.1.sta0001	04:f0:21:94:d3:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.02	0.00	71.662763	98.500169
1.1.sta0002	04:f0:21:94:f9:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.00	0.00	72.365340	97.625463
1.1.sta0003	04:f0:21:94:bd:46	11	NETGEAR_2G_wpa2	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.00	0.00	76.219512	97.704641
1.1.sta0004	00:0a:52:07:e6:c5	149	NETGEAR_5G_wpa2	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.00	0.99	0.000000	0.000000
1.1.sta0005	00:0a:52:07:fa:c5	149	NETGEAR_5G_wpa2	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.99	1.01	0.000000	0.000000
1.1.sta0006	00:0a:52:07:aa:c5	149	NETGEAR_5G_wpa2	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.99	0.99	0.000000	0.000000
1.1.sta0007	00:0a:52:07:7e:c5	149	NETGEAR_5G_wpa2	Background	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.00	1.00	0.000000	0.000000

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 8 clients running BE (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

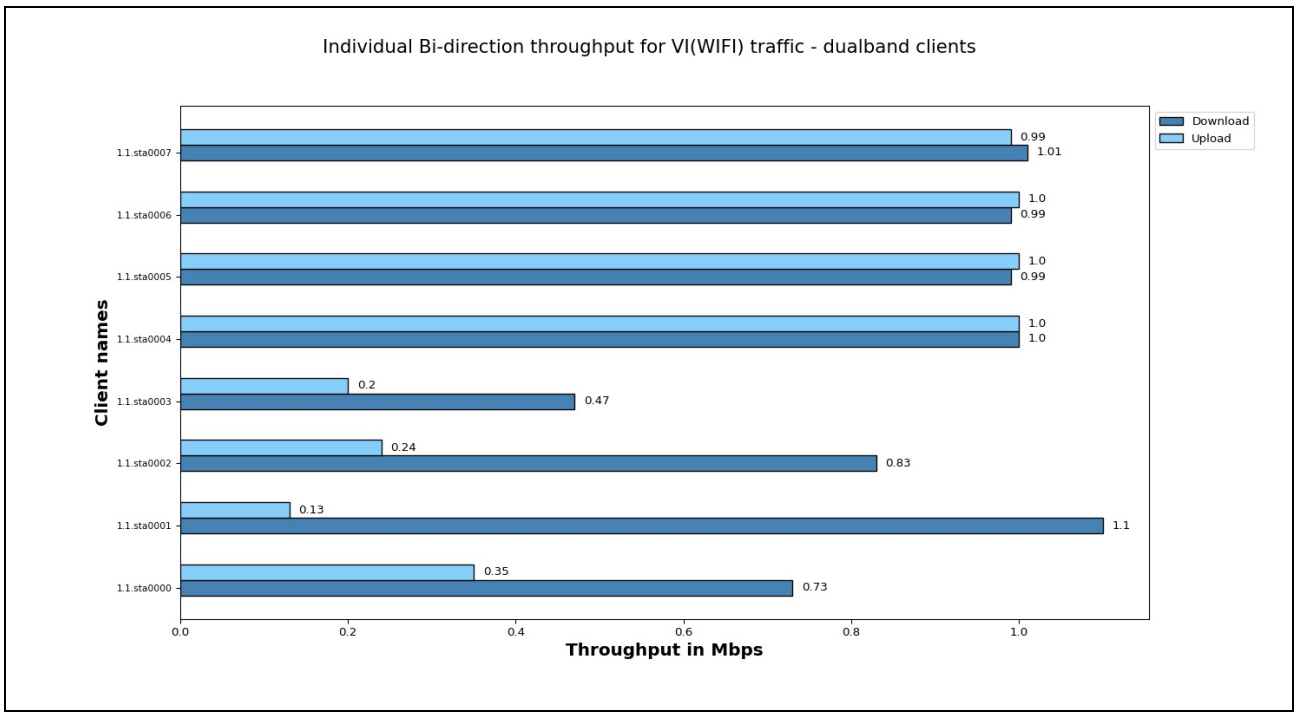


TOS : Besteffort

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:de:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.02	0.58	79.045307	78.377918
1.1.sta0001	04:f0:21:94:d3:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.08	0.47	82.119816	81.923602
1.1.sta0002	04:f0:21:94:f9:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.06	0.03	80.397727	80.743872
1.1.sta0003	04:f0:21:94:bd:46	11	NETGEAR_2G_wpa2	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.00	0.43	84.073107	77.890983
1.1.sta0004	00:0a:52:07:e6:c5	149	NETGEAR_5G_wpa2	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.99	0.99	0.000000	0.000000
1.1.sta0005	00:0a:52:07:fa:c5	149	NETGEAR_5G_wpa2	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.01	1.01	0.000000	0.000000
1.1.sta0006	00:0a:52:07:aa:c5	149	NETGEAR_5G_wpa2	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.99	1.00	0.000000	0.000000
1.1.sta0007	00:0a:52:07:7e:c5	149	NETGEAR_5G_wpa2	Besteffort	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.01	1.00	0.000000	0.000000

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VI(WiFi).

The below graph represents individual throughput for 8 clients running VI (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

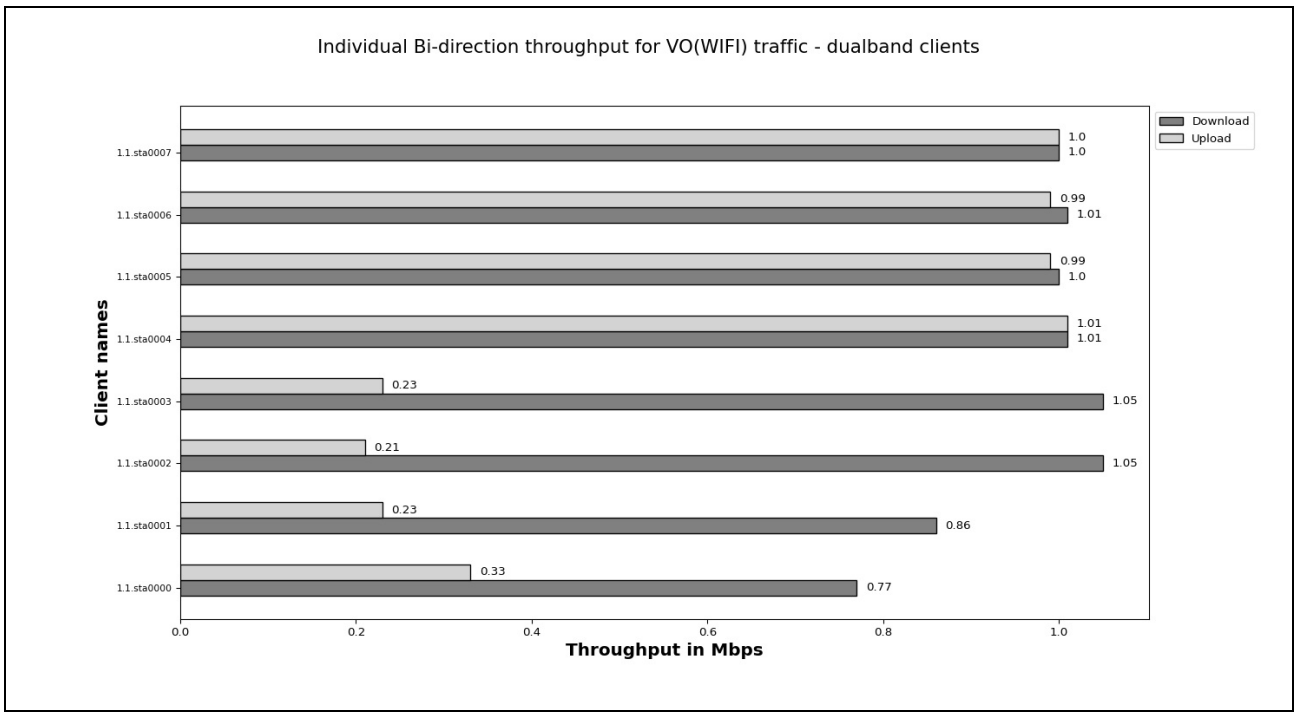


TOS : Video

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:de:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.35	0.73	74.123059	45.228287
1.1.sta0001	04:f0:21:94:d3:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.13	1.10	81.594968	43.193851
1.1.sta0002	04:f0:21:94:f9:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.24	0.83	82.251438	37.863248
1.1.sta0003	04:f0:21:94:bd:46	11	NETGEAR_2G_wpa2	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.20	0.47	83.025977	41.105029
1.1.sta0004	00:0a:52:07:e6:c5	149	NETGEAR_5G_wpa2	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.00	1.00	0.016835	0.000000
1.1.sta0005	00:0a:52:07:fa:c5	149	NETGEAR_5G_wpa2	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.00	0.99	0.000000	0.000000
1.1.sta0006	00:0a:52:07:aa:c5	149	NETGEAR_5G_wpa2	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.00	0.99	0.000000	0.000000
1.1.sta0007	00:0a:52:07:7e:c5	149	NETGEAR_5G_wpa2	Video	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.99	1.01	0.000000	0.000000

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 8 clients running VO (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.



TOS : Voice

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Upload Drop (%)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:de:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.33	0.77	76.441572	19.093407
1.1.sta0001	04:f0:21:94:d3:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.23	0.86	81.954240	20.702117
1.1.sta0002	04:f0:21:94:f9:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.21	1.05	80.725547	18.377540
1.1.sta0003	04:f0:21:94:bd:46	11	NETGEAR_2G_wpa2	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.23	1.05	81.177868	23.832500
1.1.sta0004	00:0a:52:07:e6:c5	149	NETGEAR_5G_wpa2	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.01	1.01	0.000000	0.000000
1.1.sta0005	00:0a:52:07:fa:c5	149	NETGEAR_5G_wpa2	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.99	1.00	0.000000	0.000000
1.1.sta0006	00:0a:52:07:aa:c5	149	NETGEAR_5G_wpa2	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	0.99	1.01	0.000000	0.000000
1.1.sta0007	00:0a:52:07:7e:c5	149	NETGEAR_5G_wpa2	Voice	Bi-direction	UDP	1.0 Mbps	1.0 Mbps	1.00	1.00	0.000000	0.000000

Information	contact	support@candelatech.com